

ERASTEEL TECHNICAL DATASHEET: ASP 2053

Application Family: Specialties | Standards: Erasteel High Vanadium Cold & Specialty PM

1. Grade Overview & Characteristics

ASP 2053 is a non-cobalt PM specialty grade with 8.0% Vanadium, developed for maximum abrasive wear resistance without the cost and brittleness of cobalt additions. Outstanding for plastic and composite processing.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	2.40 - 2.56	4.0 - 4.4	3.0 - 3.2	4.0 - 4.4	7.8 - 8.2	0.0	8% Vanadium, cobalt-free

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	64.0 - 66.0 HRC	Machinability	4.5/10
Max Reachable Hardness	66.0 HRC	Wear Resistance	9.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	6.0/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	6.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	4.5/10

4. Industrial Applications

Wear parts in plastic compounding, feed screws, pelletizing rings, shredder blades, cold punches for glass-fiber filled plastics.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1100-1160°C. Triple tempering at 560°C. Maximum working hardness 66 HRC.